

1 Seasonal Patterns in Aquatic Microbial 2 Communities Across Freshwater Ecosystems

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6 ABSTRACT

7 Seasonal variation is a key driver of microbial community structure in freshwater
8 ecosystems, influencing diversity, composition, and ecosystem function. However, the
9 consistency of seasonal microbial turnover across years and between nearby freshwater
10 systems remains unclear. In this study, we used National Ecological Observatory Network
11 (NEON) surface water microbial community taxonomy data (DP1.20141.002) from two
12 Virginia streams, Posey Creek (POSE) and Lewis Run (LEWI), to evaluate seasonal
13 patterns in alpha diversity and multivariate community composition across multiple
14 years. We constructed genus-level community matrices and quantified diversity using
15 Shannon indices and Bray—Curtis dissimilarity. After quality filtering, the final dataset
16 included 106 samples and 3,752 genera. Alpha diversity by Shannon indices differed
17 significantly by site, year, and season, with consistently higher diversity observed at POSE
18 compared to LEWI, although seasonal differences were weak and not driven by distinct
19 pairwise contrasts. Beta diversity was analyzed using Bray—Curtis dissimilarity with
20 PERMANOVA and principal coordinate analysis. Beta diversity differed by season and
21 site, while results for year were not interpretable due to unequal dispersion. However, the
22 low community variance explained by season and site suggests that season and site only

23 account for part of the observed community variation. Together, Alpha and Beta diversity
24 suggest that freshwater microbial communities exhibit repeatable but site-dependent
25 seasonal variation, with moderate compositional shifts over time likely influenced by
26 underlying environmental gradients rather than a single dominant driver.

27 **IMPORTANCE**

28 Freshwater microbial communities regulate nutrient cycling and carbon processing and
29 can respond rapidly to changes in temperature, hydrology, and nutrient availability.
30 Determining whether microbial communities show repeatable seasonal turnover across
31 years provides evidence for ecosystem stability and improves interpretation of long-term
32 environmental monitoring datasets. By leveraging standardized NEON sampling and
33 sequence-based taxonomy across two freshwater streams, this study establishes a
34 reproducible framework for detecting seasonal microbial patterns and for comparing
35 turnover signals between sites.

36 **INTRODUCTION**

37 Seasonal variation is a major driver of aquatic microbial community structure, shaping
38 taxonomic composition, diversity, and ecological function [10, 11, 12]. Because
39 environmental conditions such as temperature, nutrient availability, light, and hydrology
40 fluctuate in predictable annual cycles, microbial communities are expected to exhibit
41 recurring seasonal patterns in community composition, often referred to as seasonal
42 microbial turnover [2, 6].

43 Temporal dynamics in microbial communities have been observed across freshwater,
44 estuarine, and terrestrial systems [3, 4, 5]. Understanding whether these patterns are
45 stable across years and consistent across sites is important for interpreting long-term
46 monitoring data and for evaluating ecosystem resilience under environmental change [1,
47 13].

48 In this study, we evaluate whether freshwater microbial communities exhibit
49 consistent seasonal turnover across years and between nearby stream ecosystems using
50 standardized NEON data [7] from two temperate freshwater streams in Virginia: Posey
51 Creek (POSE) and Lewis Run (LEWI) (Fig. 1). We specifically test (i) whether alpha
52 diversity differs by season and site and (ii) whether multivariate community composition
53 exhibits repeatable seasonal structure across years. We hypothesize that microbial
54 communities will show consistent seasonal turnover patterns across years, with distinct
55 clustering by season in multivariate ordination space.

56 **MATERIALS AND METHODS**

57 **Data Source and Study Sites**

58 Taxonomic community data were obtained from the NEON Surface Water Microbe
59 Community Taxonomy data product (DP1.20141.002) (Fig. 2). Samples originate from two
60 NEON freshwater stream sites in Virginia: Posey Creek (POSE) and Lewis Run (LEWI).
61 The water samples were analyzed for 16S rRNA to identify microbial taxa. The study
62 period spans 2019–2024, with 2018 and 2020 excluded due to incomplete sampling in the
63 available records (Fig. 3).

64 **Variables and Sampling Structure**

65 Key fields used in analyses included dnaSampleID (unique sample identifier),
66 collectDate (collection timestamp), siteID (POSE/LEWI), taxonomic ranks through genus,
67 genusConfidence (taxonomic assignment confidence), individualCount (genus count per
68 sample), and sampleFilteredReadNumber (sequencing depth metric).

69 Seasons were defined as: Winter (Dec–Feb), Spring (Mar–May), Summer (Jun–Aug),
70 and Fall (Sep–Nov). Sampling is approximately monthly across sites and years (excluding
71 2018 and 2020).

72 **Exploratory Data Analysis**

73 Initial exploratory analyses were conducted to evaluate sequencing depth, sample
74 distribution across years, and diversity patterns across sites and seasons. These analyses
75 included summary statistics, visualization of sampling effort, and preliminary ordination
76 to assess community structure prior to formal testing. All data analysis was performed in
77 RStudio [8].

78 **Data Processing and Quality Control**

79 To support reproducible analyses, the taxonomy table was filtered to remove
80 incomplete records (e.g., missing key fields) and to exclude the years 2018 and 2020.
81 Genus-level assignments with low confidence were filtered using a genusConfidence
82 threshold (values below 0.1 excluded). Samples with low sequencing depth (<5,000 reads)
83 were also removed during quality control. After filtering and reshaping, the final dataset
84 used for community analyses contained 106 water samples and 3,752 genera (Fig. 3).

85 **Community Matrix Construction**

86 Taxonomic counts were aggregated at the genus level for each unique dnaSampleID,
87 yielding one row per sample and one column per genus. The table was reshaped into a
88 wide community matrix with missing genus counts filled as zeros to indicate absence
89 within a sample.

90 **Alpha Diversity**

91 Alpha diversity was estimated using the Shannon diversity index computed per
92 sample from the genus-level community matrix. The Shannon index was used because it
93 incorporates both species richness and evenness, making it a widely used metric for
94 microbial community diversity. Diversity distributions were summarized by site and
95 season. A factorial ANOVA was used to test for differences in Shannon diversity across
96 year, site, and season, followed by Tukey's HSD post hoc comparisons.

97 **Beta Diversity**

98 Community composition between samples was quantified using Bray–Curtis
99 dissimilarity computed from genus-level abundance profiles. Bray–Curtis dissimilarity
100 was used because it is appropriate for abundance-based ecological community data and is
101 commonly applied in microbial ecology studies. Principal coordinates analysis (PCoA)
102 was used to visualize patterns in multivariate community composition [9]. A
103 PERMANOVA test was used for statistical analysis along with tests for homogeneity of
104 multivariate dispersion.

105 **RESULTS**

106 **Sampling Effort and Dataset Characteristics**

107 Samples were collected from two NEON sites (POSE and LEWI) across multiple years
108 (2018-2024). The sampling effort varied across years (Fig. 2–5), with fewer observations in
109 earlier years (2019) and increased in later years (2021-2024). Both sites were represented
110 across all included seasons, though the number of samples per season and year was not
111 uniform (Fig. 2). The samples spanned all four seasons (Winter, Spring, Summer, and Fall),
112 with multiple observations per season in most years (Fig. 4). To align with the definitions
113 of the meteorological season, samples collected in December were assigned to the
114 following year. Sequencing data were normalized before analysis to account for
115 differences in sequencing depth between samples (Table 3). Years 2018 and 2020 were
116 excluded from further analyses due to incomplete seasonal coverage and reduced sample
117 representation following data filtering and temporal adjustment (Fig. 3). The samples
118 were analyzed for seasonal changes in microbial diversity. The data analysis performed
119 includes testing Shannon diversity index (Alpha diversity) and Bray–Curtis Dissimilarity
120 (Beta diversity) by season, site, and year. Sampling effort was relatively balanced between
121 sites, with 52 samples from LEWI and 54 from POSE, and seasonal coverage included all
122 four seasons with slightly higher representation in summer compared to winter (Fig.4,
123 Fig.5).

124 **Alpha Diversity Patterns**

125 Shannon diversity values ranged from 3.86 to 5.77, with a median of 5.42. Observed
126 genus richness ranged from 192 to 1,592 genera per sample, with a median of 1,195
127 (Fig. 6).

128 A factorial ANOVA was used to test whether mean Shannon diversity differed across
129 year, site, and season. The F-statistic represents the ratio of variation between groups to
130 variation within groups, with higher values indicating stronger differences between
131 categories. Shannon diversity differed significantly by year ($F_{4,67} = 24.12$, $p < 0.001$), site
132 ($F_{1,67} = 14.97$, $p = 0.00025$), and season ($F_{3,67} = 2.88$, $p = 0.042$). The relatively high
133 F-value for year suggests that temporal variation had a stronger effect on Shannon
134 diversity than seasonal differences, while the significant site effect indicates consistent
135 differences in diversity between POSE and LEWI. There was also a significant interaction
136 between year and site ($F_{4,67} = 2.74$, $p = 0.036$), while all other interaction terms were not
137 significant ($p > 0.05$) (Table 5).

138 Post hoc comparisons using Tukey's HSD showed that Shannon diversity in 2019
139 differed significantly from all other years (2021-2024; all adjusted $p < 0.001$), while no
140 significant differences were observed between 2021-2024 (all adjusted $p > 0.05$). Although
141 season was statistically significant overall, pairwise comparisons between seasons were
142 not significant, indicating that seasonal differences in alpha diversity were subtle and not
143 driven by strong contrasts between individual seasons (all adjusted $p > 0.05$) (Fig. 7).

144 **Beta Diversity Patterns**

145 Changes in microbial community composition were assessed using Bray—Curtis
146 dissimilarity. Principal coordinate analysis (PCoA) was used to visualize patterns in
147 community structure (Fig. 8, Fig. 9). There was a low variance explained by both PCoA1
148 (28.5%) and PCoA2 (15.16%), but this is common when analyzing microbial communities
149 (Fig. 9, Table 4). There is a pattern of the winter seasons having a lower PCoA1 value and

the summer seasons having a higher PCoA1 value. The figure also shows clear clustering between the LEWI and POSE sites (Fig. 9).

PERMANOVA tests were performed to test the significance of the Bray—Curtis dissimilarity based on season, year, and site. The Bray—Curtis dissimilarity differed significantly by season ($F_{3,102} = 7.06$, $R^2 = 0.172$, $p < 0.001$), site ($F_{1,104} = 23.34$, $R^2 = 0.183$, $p < 0.001$) and year ($F_{4,101} = 4.41$, $R^2 = 0.149$, $p < 0.001$). However, the explained variance (R^2) for Beta diversity is low, indicating that season, site, and year explain only a portion of the variation in microbial community composition. (Table 4).

Tests for homogeneity of multivariate dispersion showed no significant differences among seasons ($F_{3,102} = 2.31$, $p = 0.081$) or site ($F_{1,104} = 2.09$, $p = 0.151$) indicating that PERMANOVA results were not driven by unequal dispersion. Tests for homogeneity of multivariate dispersion between years showed significant differences ($F_{4,101} = 5.23$, $p = 0.0007$) indicating PERMANOVA results were driven by unequal differences.

DISCUSSION

Freshwater microbial communities show predictable temporal variation that shapes patterns of diversity, composition, and ecosystem function [10, 11]. In this study, we evaluated whether freshwater communities exhibit consistent seasonal turnover across years and between two stream sites using NEON taxonomy data. Shannon diversity varied by site and season, with clear differences in both magnitude and variability. Overall, POSE exhibited higher diversity than LEWI, indicating greater community evenness and richness (Fig. 6, Fig. 7). Seasonal patterns were observed at both sites, with slightly higher diversity during spring and summer, particularly at POSE. In contrast, LEWI showed greater variability across seasons, including lower diversity values during fall and winter, suggesting periods of increased dominance by fewer taxa. These patterns indicate that while both streams experience seasonal influences, POSE maintains a more consistently diverse and evenly distributed community, whereas LEWI exhibits greater fluctuations in evenness and community stability across seasons.

177 However, although seasonal effects on alpha diversity were statistically significant,
178 post hoc comparisons did not identify significant differences between specific seasons,
179 indicating that these effects are relatively weak and not driven by distinct seasonal
180 contrasts. Instead, differences in diversity appear to be more strongly influenced by site
181 level factors, with LEWI exhibiting greater variability and occasional dominance by fewer
182 taxa. This suggests that spatial differences between streams may play a larger role in
183 shaping microbial diversity than seasonal variation alone.

184 Analysis of community composition using Bray—Curtis dissimilarity showed
185 significant differences by site and season (Table 4), however the proportion of explained
186 variance was relatively low, indicating that these factors account for only a portion of the
187 observed variation in diversity. The Bray—Curtis dissimilarity results between years were
188 unreliable due to PERMANOVA results being driven by unequal dispersion (Table 4).
189 PCoA further supports these findings, showing moderate seasonal clustering with
190 overlap between seasons, as well as clear separation between POSE and LEWI (Fig. 8,
191 Fig. 9). While seasonal clustering was observed, it was moderate rather than distinct,
192 indicating that seasonal turnover occurs along a continuum rather than forming clearly
193 separated seasonal community groups [11].

194 These patterns are consistent with the distribution of dominant taxa observed in the
195 community composition data. At LEWI, communities were frequently dominated by a
196 single genus, particularly *Rhodoferrax*, with a sharp decline in relative abundance across
197 ranks, indicating low evenness. In contrast, POSE displayed a more even distribution of
198 taxa, with multiple genera contributing similar relative abundances rather than a single
199 dominant taxon (Table 6). This greater evenness at POSE likely explains the higher
200 Shannon diversity observed across seasons, consistent with established relationships
201 between evenness and diversity in microbial communities [11].

202 Overall, our findings are consistent with previous studies showing that freshwater
203 microbial communities vary over time and across locations [2, 4, 10]. Prior work has
204 linked these patterns to seasonal changes in environmental conditions such as
205 temperature, hydrology, and nutrient availability [6, 10, 12]. Similar trends were observed

206 here, although patterns were not strongly separated, suggesting that microbial dynamics
207 are driven by multiple interacting environmental factors rather than a single dominant
208 driver, consistent with prior work highlighting the complexity of microbial–environment
209 interactions [14]. In this context, season should be interpreted as a proxy for underlying
210 environmental gradients, including temperature, nutrient availability, hydrology, and
211 water chemistry, which were not directly measured in this study but are known to
212 influence microbial community structure [6, 12].

213 It is important to note that season and site likely reflect underlying environmental
214 gradients that are not directly measured in this study. Factors such as water chemistry,
215 nutrient availability, and stream flow may be helping to shape microbial community
216 structure and could explain why some patterns were present but not strongly distinct [10,
217 12]. These results suggest that freshwater microbial communities exhibit repeatable but
218 site-dependent seasonal variation, characterized by moderate shifts in community
219 composition driven primarily by changes in relative abundance rather than the
220 appearance or disappearance of taxa [11].

221 There are a few limitations to consider. Sampling was not evenly distributed across
222 years, and some years had reduced coverage after filtering, which may have influenced
223 the strength of the patterns observed. In addition, using genus-level classifications from
224 16S rRNA data provides a general view of community structure but does not capture
225 more detailed taxonomic or functional differences. Lastly, although the statistical tests
226 identified significant effects, the overall explained variation was moderate, suggesting
227 that other unmeasured environmental factors likely contribute to microbial community
228 dynamics [10].

229 Future research could expand on this work by including additional freshwater sites to
230 determine whether the patterns observed here are consistent across a broader range of
231 ecosystems. Incorporating other microbial groups, such as fungal communities, would
232 provide a more comprehensive view of microbial dynamics and ecosystem function.
233 Further investigation into the environmental factors driving seasonal dominance of
234 specific taxa would help clarify the ecological mechanisms underlying these patterns.

235 Extending analyses across longer time scales would also improve the ability to detect
236 consistent interannual trends and identify potential early signals of climate-related
237 changes influencing microbial community structure.

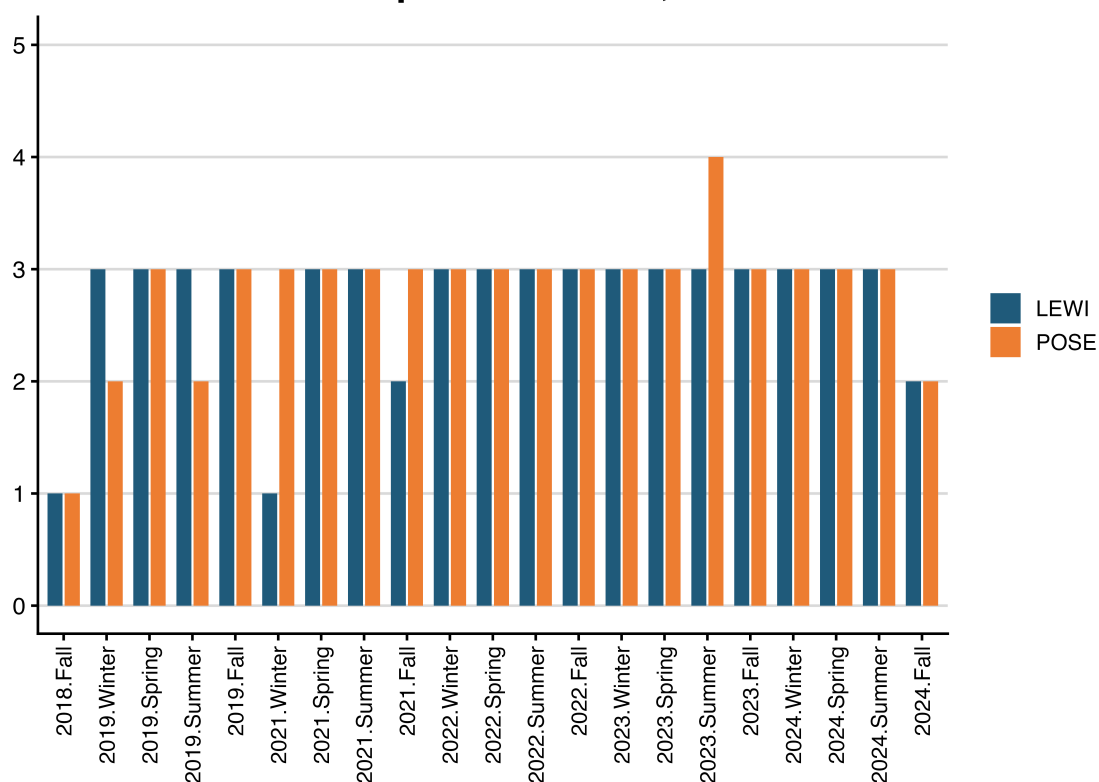
238 **Figures**

239 The NEON sampling locations used in this study are shown in Fig. 1. Sampling effort
240 and dataset composition are summarized in Fig. 2–5, including the distribution of
241 samples across years, seasons, and study sites following filtering and quality control.
242 Alpha diversity patterns, including observed genus richness and Shannon diversity by
243 season and site, are presented in Fig. 6 and Fig. 7. Community composition is visualized
244 using principal coordinates analysis (PCoA) of Bray—Curtis dissimilarity, highlighting
245 patterns of seasonal clustering and site-level differences (Fig. 8 and Fig. 9).



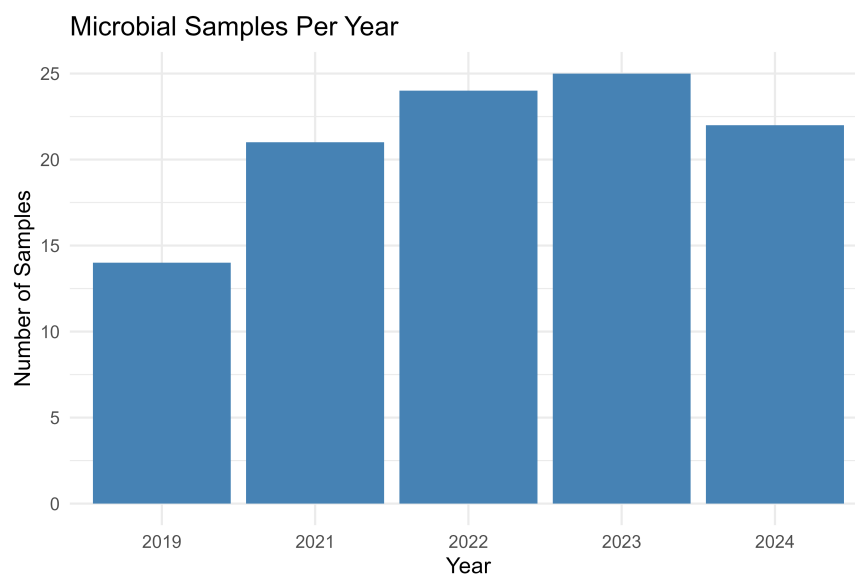
FIG 1 Study sites included in this analysis: Posey Creek (POSE) and Lewis Run (LEWI), Virginia, USA. Samples were obtained from the National Ecological Observatory Network (NEON) Surface Water Microbe Community Taxonomy dataset (DP1.20141.002). Map adapted from the NEON field sites and ecoclimatic domains.

Number of Samples Collected, 2019–2024



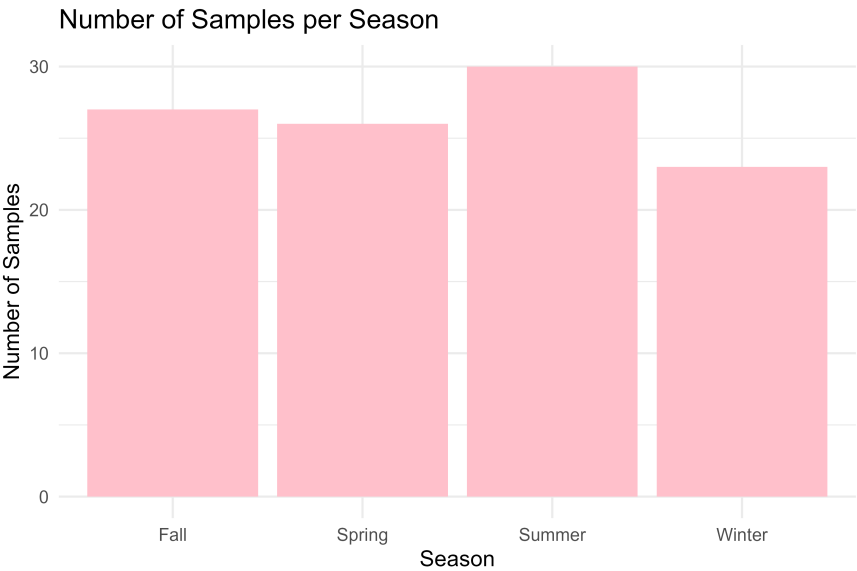
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FIG 2 All available microbial community samples collected per year at the POSE and LEWI stream sites between 2018 and 2024. Bars show the number of samples per site per year used in the community analysis dataset.



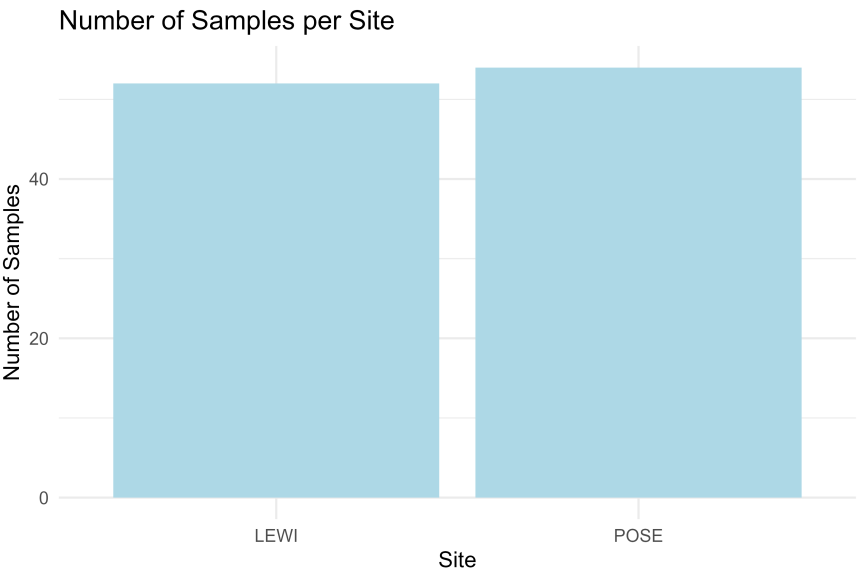
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FIG 3 Number of microbial samples per year included in the final dataset after quality filtering and temporal adjustment. Sample counts reflect observations from both POSE and LEWI sites.



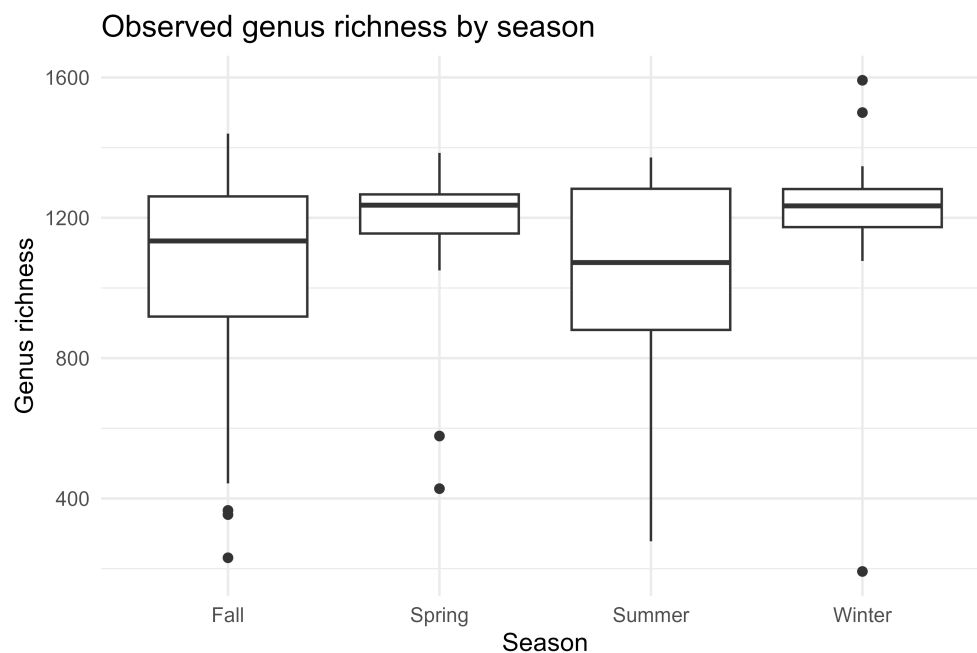
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FIG 4 Distribution of microbial samples across seasons (Fall, Spring, Summer, Winter) following filtering and exclusion of incomplete years. Sample counts reflect observations from both POSE and LEWI sites.



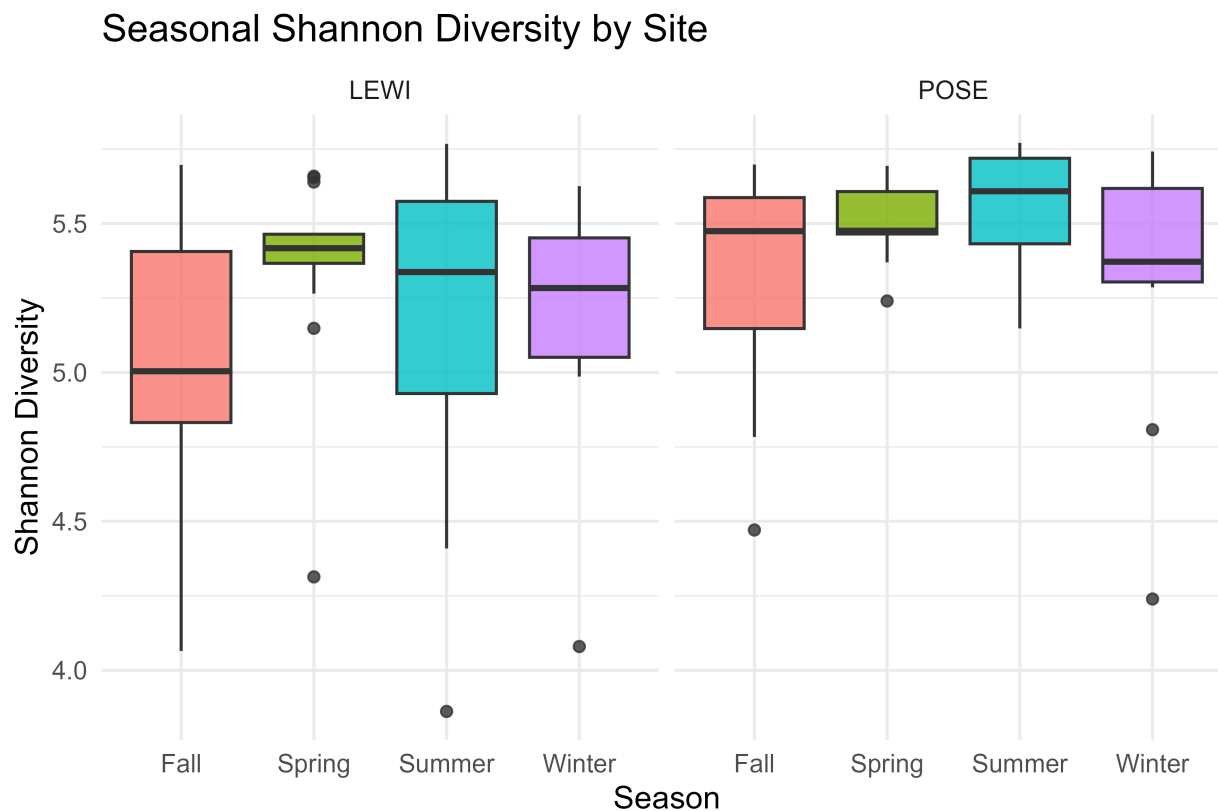
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FIG 5 Distribution of microbial samples across study sites (LEWI and POSE) following filtering and exclusion of incomplete years. Sample counts indicate comparable representation between sites.



251

FIG 6 Observed genus richness by season.



252

FIG 7 Seasonal Shannon diversity across sites. Shannon diversity index values were calculated from genus-level community matrices derived from NEON surface water microbial samples collected between 2019 and 2021–2024 (2018 and 2020 excluded). Boxplots summarize diversity distributions by season for each site (LEWI and POSE).

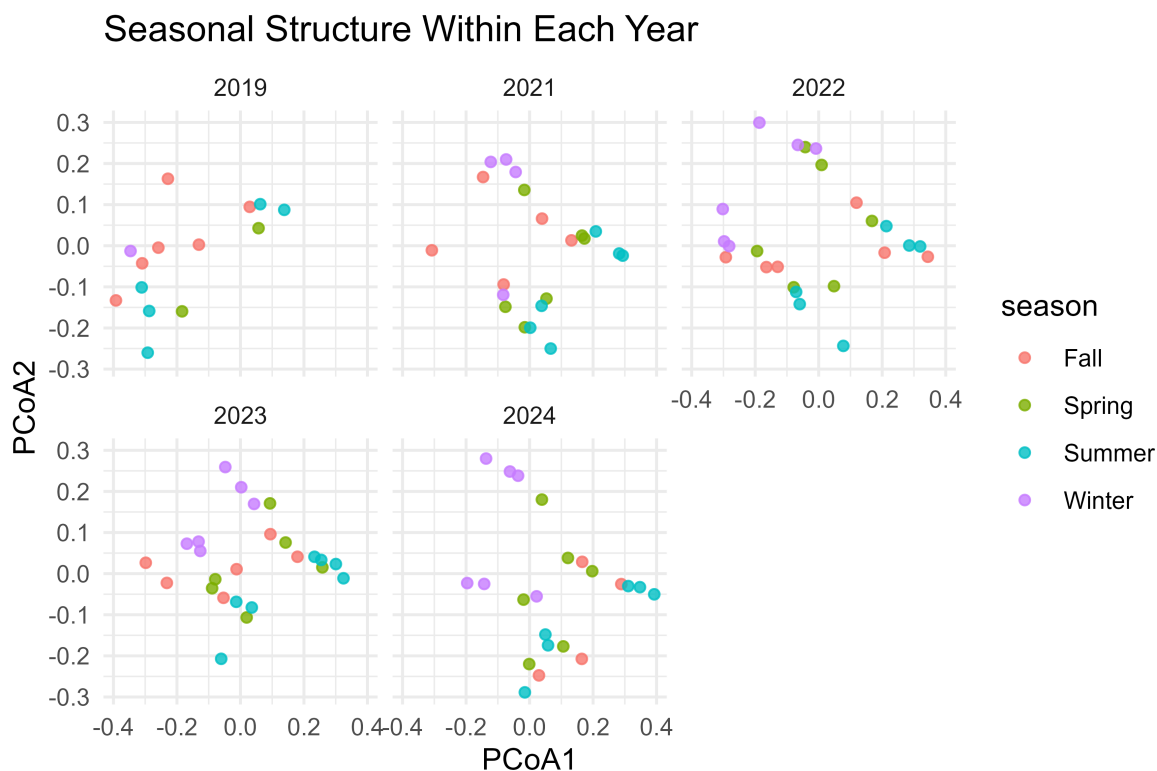


FIG 8 Principal coordinates analysis (PCoA) of Bray—Curtis dissimilarities for genus-level microbial community composition. Points represent individual samples and are colored by season to visualize seasonal clustering in community structure. Axes represent the first two principal coordinates (PCoA1 and PCoA2).

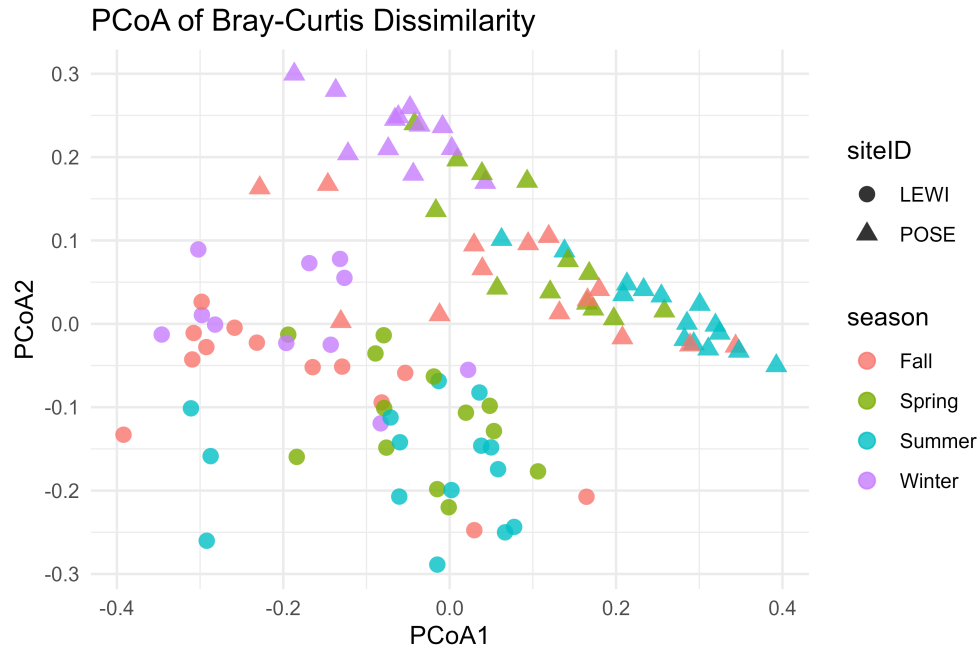


FIG 9 Principal coordinates analysis (PCoA) of Bray–Curtis dissimilarities for genus-level microbial community composition. Points represent individual samples; colors indicate season (fall, spring, summer, winter) and shapes indicate site (LEWI, POSE). Axes show the first two principal coordinates (PCoA1 and PCoA2).

255 Tables

256 Tables 1 and 2 summarize the analytical framework and sampling structure.
 257 Additional sequencing depth, ordination variance, and diversity summaries are provided
 258 in Tables 3–5. Dominant genus patterns by site and year across seasons are presented in
 259 Table 6.

TABLE 1 Study variables and analytical framework.

260

Variable	Definition	Type	Analytical method
Site	POSE, LEWI	Categorical	Grouping factor
Year	2019, 2021–2024	Categorical	Fixed factor
Season	Winter, Spring, Summer, Fall	Categorical	Grouping factor
Alpha diversity	Within-sample diversity	Continuous	Shannon diversity index
Community composition	Genus-level abundance profile	Multivariate	Bray–Curtis dissimilarity
Ordination	Multivariate visualization	Multivariate	PCoA
Seasonal difference	Seasonal community differences	Multivariate	PERMANOVA

TABLE 2 Sampling structure summary for NEON surface water microbial samples included in analyses (2019 and 2021–2024; 2018 and 2020 excluded).

261

Site	Years included	Sampling frequency
POSE	2019, 2021–2024	Approximately monthly
LEWI	2019, 2021–2024	Approximately monthly

TABLE 3 Summary of sequencing depth (total reads per sample) after filtering low-read samples (<5,000 reads) across all included samples (2019–2024, excluding 2018 and 2020).

262

Minimum	Median	Mean	Maximum
5290	218972	210393	944705

TABLE 4 Percent variance explained by principal coordinates.

263

Axis	Percent variance explained
PCoA1	28.56
PCoA2	15.16

TABLE 5 Sequencing depth and alpha diversity summary by site (2019–2024, excluding 2018 and 2020).

Site ID	Samples (n)	Mean reads	Median reads	Mean Shannon	SD Shannon	Mean richness
LEWI	52	203904.4	206478	5.19	0.46	1085
POSE	54	216642.1	225903	5.44	0.312	1102.8

TABLE 6 Dominant genus by site and year across seasons.

siteID	year_adj	Fall	Spring	Summer	Winter
LEWI	2019	Rhodoferax	Flavobacterium	Rhodoferax	Rhodoferax
LEWI	2021	Rhodoferax	Rhodoferax	Rhodoferax	Rhodoferax
LEWI	2022	Rhodoferax	Rhodoferax	Cyanobium	Rhodoferax
LEWI	2023	Rhodoferax	Bacillariophyta	Rhodoferax	Bacillariophyta
LEWI	2024	Rhodoferax	Rhodoferax	Rhodoferax	Rhodoferax
POSE	2019	Rhodoferax	Rhodoferax	Bacillariophyta	NA
POSE	2021	Rhodoferax	Rhodoferax	Gp6	Flavobacterium
POSE	2022	Gp6	Bacillariophyta	Gp6	Bacillariophyta
POSE	2023	Rhodoferax	Bacillariophyta	Gp6	Bacillariophyta
POSE	2024	Rhodoferax	Rhodoferax	Gp6	Flavobacterium

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